M2 Research internship proposal

Bayesian segmentation models for polarimetric images: application in biological imaging

supervision: J. Flamant (CRAN) and R. Stoica (IECL)

Internship research project Imaging techniques that exploits polarization properties of light are becoming increasingly popular in biology, as polarimetric imaging enables many insights that are inaccessible to other conventional imaging methods [1]. As an example, Fig. 1 illustrates some of the benefits of polarimetric measurements in bio-imaging: compared to conventional, intensity-only imaging polarization information provides crucial physical and morphological insights together with increased image contrast.

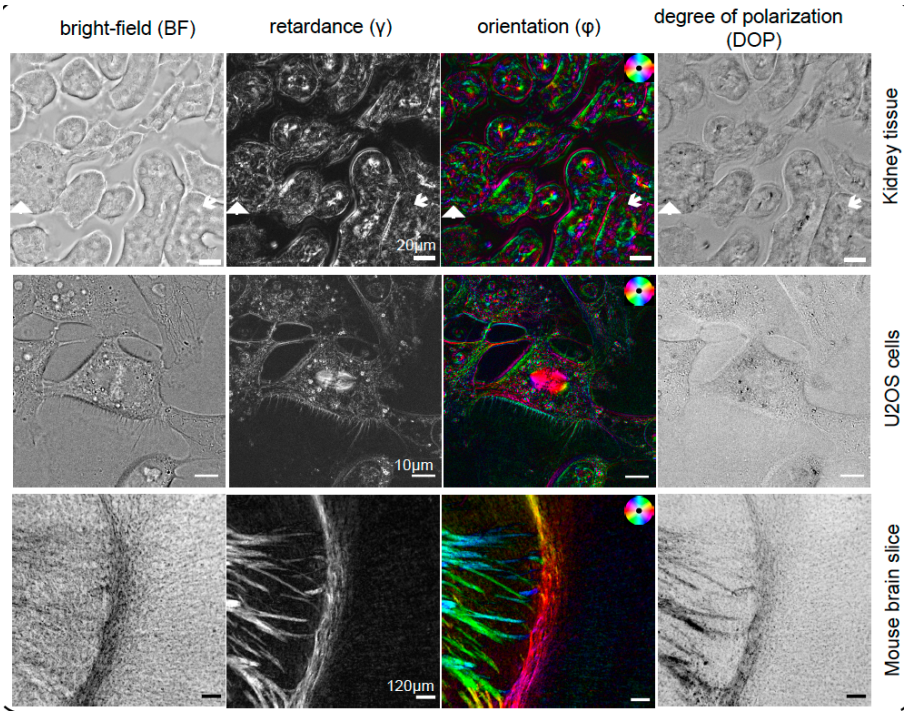

Figure 1: Adapted from [2]. Rowwise: three different biological scenes. Columnwise: standard intensity bright field image, the retardation due to the birefringence of the medium, the orientation of the optical axis and the degree of polarization. These last three quantities can only be obtained by measuring polarization.

Polarization information is typically described using Stokes parameters, a set of four energetic parameters that can need to be estimated from multiple intensity noisy measurements. This internship will focus on the segmentation problem of Stokes images and develop efficient algorithms for its resolution. A first task will consist in formulating the segmentation problem in a Bayesian framework

with relevant data-fidelity terms for two noise models (additive Gaussian, Poisson) as well as spatial regularization thanks to Markov Random Fields (MRF) [3]. The candidate will develop Markov-Chain Monte Carlo (MCMC) algorithms to solve the polarimetric segmentation problem, starting with the supervised case. The unsupervised polarimetric segmentation will be considered as a second step, focusing on developing efficient parameter estimation methods [4] that accounts for physical constraints associated with Stokes parameters [5], [6].

The second aim of this internship will be the application of the proposed methodology to cell biological imaging. In particular, one goal will be to investigate the potential of polarimetric measurements for characterizing cellular senescence. This work will be done in close collaboration with cell biologists from CRAN. This internship can be followed by a PhD position depending on the results and motivation of the candidate

Prerequisites and funding The candidate should be enrolled in a M2 and have a M1 in either mathematics, applied mathematics or electrical engineering (signal processing), together with good oral and written english skills. The internship offers also a grant (around 550 euros per month w.r.t. the French rules). The internship is 6 months length, and it may start anytime during the period of 01/02 - 01/04/2021.

Location The candidate will be working mainly at the Centre de Recherche en Automatique de Nancy (CRAN), located on the Faculté des Sciences et technologies campus in Vandoeuvre-lès-Nancy, France.

Supervision This internship falls within a collaboration between CRAN and Institut Elie Cartan de Lorraine (IECL campus de Nancy) which are located a few hundred meters apart on the campus. The candidate will be jointly supervised by researchers of both institutions:

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References

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